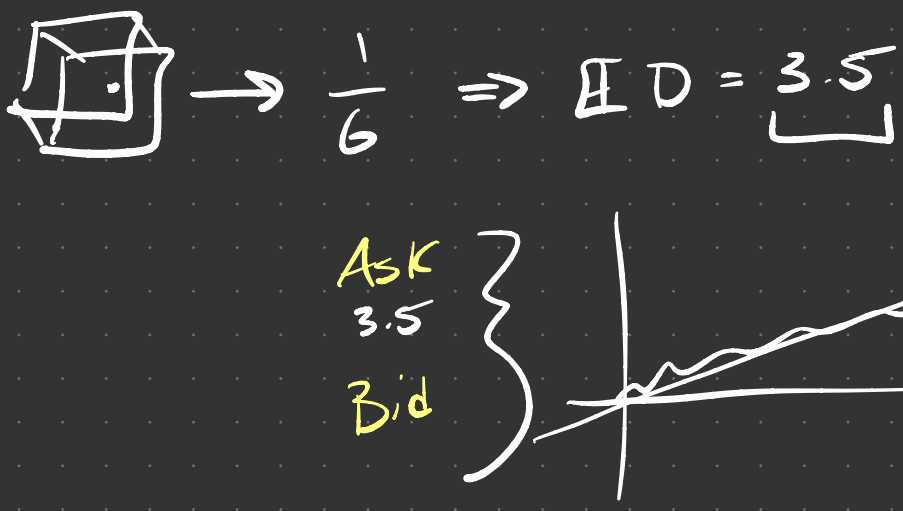


Defn: Random Experiment

A procedure or process whos outcome can't be predicted

- Coin
- Dice roll
- Roulette

Defn: Probability Model

A probability model is a math description of a Random experiment that specifies outcomes and assigns probabilities to different "events"

- What outcomes are possible?
- How likely is each event?

- How does knowing one event impact the prob. of another
- What is the exp. value of a random qty?

Defn: Sample Space

The set of all possible outcomes of a Random experiment

e.x.) Sample Space Dice Roll

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\Omega^2 = \Omega \otimes \Omega = \{(1,1), (1,2), \ldots$$
$$(2,1), (2,2), \ldots$$
$$\vdots$$
$$(6,1), (6,2), \ldots\}$$

Defn: Event

An event is a subset of the sample space.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

"Even Dice Rolls" $= E = \{2, 4, 6\}$

$$E \subseteq \Omega$$

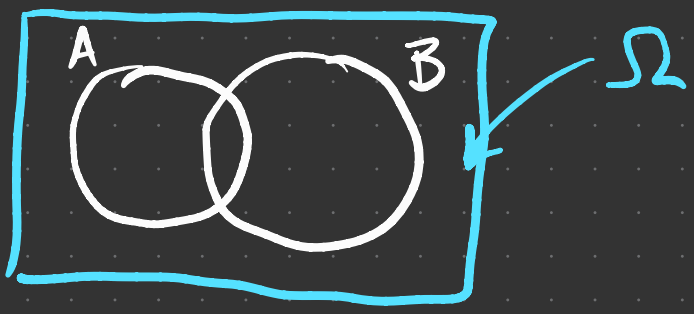
Defn: Operations on Sets

Given two events $A, B \subseteq \Omega$

Union: $A \cup B = U$

Intersection: $A \cap B = I$

Complement: $A^c = \Omega \setminus A$



Defn: Probability Measure

A probability measure, P, is a function that assign a number to each event and it satisfies the following axioms:

Axiom 1: Nonnegativity

$$P(A) \geq 0$$

Axiom 2: Normalization

$$P(\Omega) = 1$$

Axiom 3: Countable Additivity

pairwise mutually exclusive events

$\forall i, j \quad A_i \cap A_j = \emptyset$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

$$0 \leq P(A) \leq 1 \quad , \quad \forall A$$

Rule 1: Complement Rule

$$P(A^c) = 1 - P(A)$$

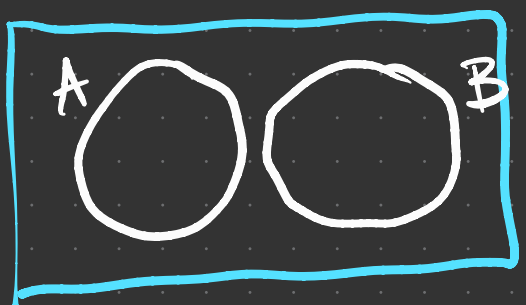
$$P(A^c) + P(A) = 1$$

$$A^c \cup A = \Omega$$

Rule 2: Addition Rule

For any two events A, B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$P(A \cap B) = 0$

$A \cap B = \emptyset$

Defn) Equally Likely Outcomes

A sample space has equally likely outcomes if every event has the same probability.

Ex) N outcomes, each has prob. $\frac{1}{N}$

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{The total \# of outcomes}}$$

E - "Even dice roll" $= \{2, 4, 6\}$

$$P(E) = \frac{|E|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}$$

Ex) Roll two dice independently.

S - "At least one six"; $P(S) = \frac{|S|}{|\Omega|} = \frac{11}{36}$

$|\Omega| = 36$, $|S| = 11$

$$\frac{25}{36} = P(S^c) \Rightarrow P(S) = 1 - P(S^c)$$

$$= 1 - \frac{25}{36} = \frac{11}{36}$$

Combinations + Permutations

Permutation Formula

$$P(n,r) = \frac{n!}{(n-r)!} \qquad \begin{matrix} n=6 \\ r=3 \end{matrix}$$

$$\frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1}$$

Combinations Formula

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

A B C D E F

6

$${}_6C_3 = \binom{6}{3} = \frac{6!}{3!(3)!} = \frac{120}{3!}$$

$$= \frac{120}{6} = 20$$

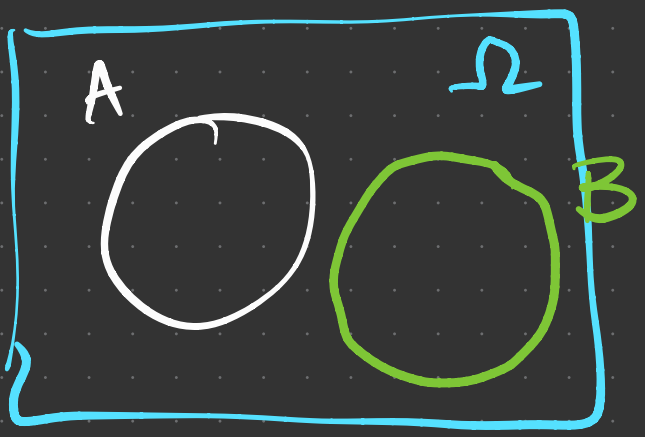
Defn: Conditional Probability

Given events A, B

The cond. prob A given B occured

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) > 0$$



Multiplication Rule

$$P(A \cap B) = P(A|B)P(B)$$

Defn: Partition

A collection of events $B_1, B_2, \ldots$ forms a partition if

1) $\forall i,j \; B_i \cap B_j = \emptyset$

2) $\bigcup_{i=1}^{\infty} B_i = \Omega$

Law of Total Probability

If $B_1, B_2, \ldots$ form a partition

$$P(A) = \sum_{i=1}^{\infty} P(A \mid B_i) P(B_i)$$

Bayes' Theorem

For a partition $B_1, B_2, \ldots$

$$P(B_i \mid A) = \frac{P(A \mid B_i) P(B_i)}{\sum_{j=1}^{\infty} P(A \mid B_j) P(B_j)}$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \mid B) P(B)}{P(A)}$$

Defn: Independent Events

Given A and B, the events are independent ($A \perp B$) if

$$P(A \cap B) = P(A) P(B)$$

Given $A \perp B$, $P(A|B) = ?$

$$P(A|B) = \frac{P(A)P(B)}{P(B)}$$

$$\Rightarrow P(A|B) = P(A) \text{ if } A \perp B$$

Defn Mutual Exclusivity

Given events A and B,
the events are mutually exclusive
if $A \cap B = \emptyset$ or $P(A \cap B) = 0$

ex.) Given $A \cap B = \emptyset$

$$\Rightarrow P(A \cap B) = 0$$

$$\Rightarrow P(A)P(B) > 0$$

If A, B ME $\Rightarrow$ Not Independent

$P \Rightarrow Q$

Defn Mutual Independence

A collection of events $A_1, A_2, \ldots, A_n$
is mutually independent if for
every non-negative subset

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = \prod_{j=1}^{k} P(A_{i_j})$$

Defn: Random Variable

A random variable (RV) is a real-valued function defined on the outcomes of a random experiment

$$X : \Omega \to \mathbb{R} \quad \text{or} \quad X : S \to \mathbb{R}$$

sample space

$$\Omega = \{HH, TH, HT, TT\}$$

Outcome (ω)	X
HH	2
TH	1
HT	1
TT	0

$$P(X=0) = \frac{1}{4}, \quad P(X=2) = \frac{1}{4}$$

$$P(X=1) = \frac{2}{4} = \frac{1}{2}$$

Defn: Discrete Random Variable

A discrete RV takes values in a finite or countably infinite set.

Defn: Prob. Mass Function (PMF)

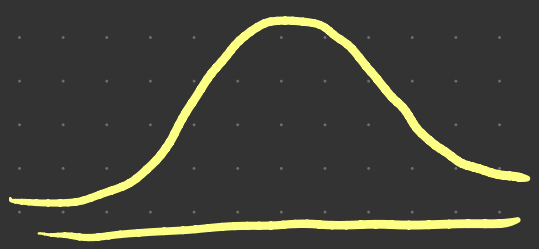
$$P_X(x) = P(X = x)$$

$$P_X(x) \geq 0$$

$$\sum_{a \in A} P_X(a) = 1$$

Defn: Continuous Random Variable

A continuous RV has a probability distribution that can be described by a probability density function (PDF)



Defn: Prob. Density Function

A function $f_X(x)$ is a PDF if

$$f_X(x) \geq 0$$

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

$$\Rightarrow P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

Defn: Cumulative Distribution Function

The CDF of a random variable X is

$$F_X(x) = P(X \leq x)$$

Cont. CDF

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$$

Essential Probability Distributions

Distribution	PMF / PDF	Mean	Variance
$X \sim Ber(p)$	$P(X=1) = p$	p	$p(1-p)$
$X \sim Bin(n,p)$	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
$X \sim Geo(p)$	$p(1-p)^{k-1}, k \geq 1$	$\frac{1}{p}$	$\frac{(1-p)}{p^2}$
$X \sim Pois(\lambda)$	$\frac{e^{-\lambda}\lambda^k}{k!}$	λ	λ
$X \sim U(a,b)$	$\frac{1}{b-a}, (a,b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$X \sim N(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2

Defn: Expected Value

The expected value of a RV X is its probability weighted sum of outcomes.

Discrete RV: $\mathbb{E}[X] = \sum_x x \, p_X(x)$

Contin. RV: $\mathbb{E}[X] = \int_{-\infty}^{\infty} t \, f_X(t) \, dt$

Defn: Variance

$$\text{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right]$$

$$= \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Defn: SD

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Properties of $\mathbb{E}$, Var

RV X; $a, b \in \mathbb{R}$

$$\mathbb{E}[aX+b] = \mathbb{E}[aX] + \mathbb{E}[b]$$
$$= a\mathbb{E}[X] + b$$

$$\mathrm{Var}(aX+b) = \mathrm{Var}(aX) + \mathrm{Var}(b)$$
$$= \mathrm{Var}(aX) + 0$$
$$= a^2 \mathrm{Var}(X)$$

Given X, Y.

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

by linearity of $\mathbb{E}$, regardless if $X \perp Y$

Defn: Covariance

Given RVs X, Y

The covariance measures linear co-variation

$$Cov(X,Y) = \mathbb{E}\left[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\right]$$

$$= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Given X, Y if we observe the var.

$$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$$

Defn: Joint Prob. Distribution

Describe the probabilities associated with two or more RVs simultaneously

For discrete X, Y the joint PMF

$$P_{X,Y}(x,y) = P(X=x, Y=y)$$

Defn: Marginal Distribution

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

Defn: Independent Random Variables

Given X, Y if the joint distribution is a product of the marginals, $X \perp Y$

$$P_{X,Y}(x,y) = P_X(x) P_Y(y)$$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

Defns LLN

Let $X_1, X_2, \ldots \overset{iid}{\sim}$.

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

The weak LLN says

$$\bar{X}_n \xrightarrow{P} \mu \quad \text{as} \quad n \to \infty$$

Defns CLT

Let $X_1, \ldots, X_n$ i.i.d random variables with μ, σ^2

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1)$$

as $n \to \infty$

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Ex) Game, where each round is independent.

$P(W) = .6$

$P(L) = .4$

What is the prob. of winning twice in a row?

$$P(WW) = .6(.6) = .36$$

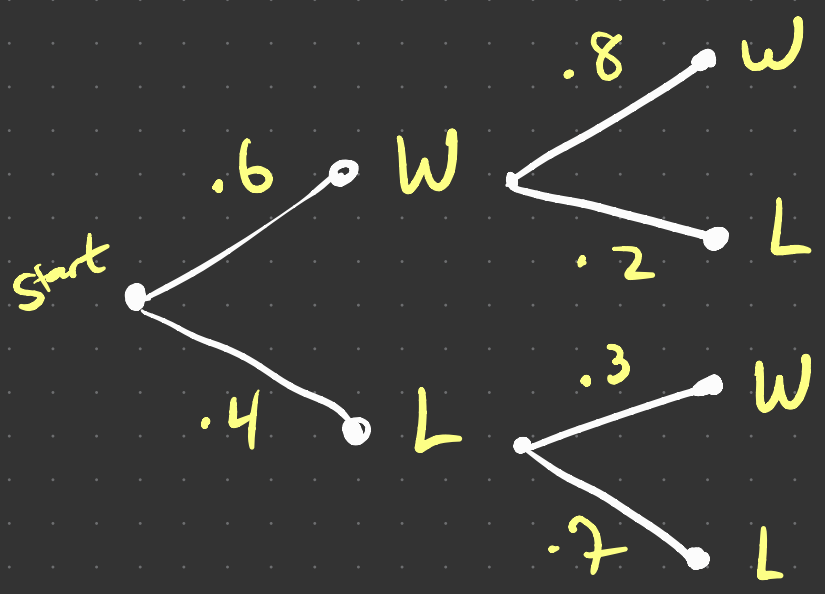
Q: What if the second round depends on the first?

Q: You're given a fair die, the game says to roll the die and receive $ amount of the face value. You can choose to reroll one time. What is the max you should be willing to pay to play?

Suppose the probability of winning the next round is given by

$$P(W_{n+1} | W_n) = .8$$

$$P(W_{n+1} \mid L_n) = .3$$



$$P(W_2) = P(W_2 \mid W_1)P(W_1) + P(W_2 \mid L_1)P(L_1)$$

$$= (.8)(.6) + (.3)(.4)$$

$$= .48 + .12 = .6$$

$$P(W_2) = .6$$

First Step Analysis

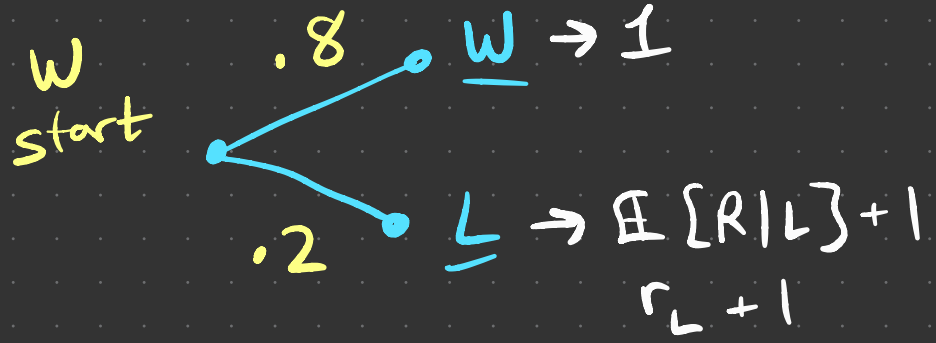
Given a RV Y that is integrable

$$\mathbb{E}[Y] = \sum_i P(X_1 = i)\,\mathbb{E}[Y \mid X_1 = i]$$

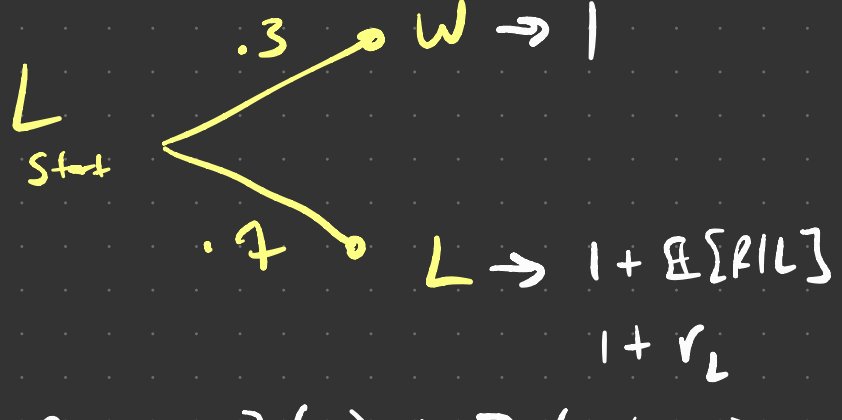
Q: Starting immediately after a win, what is the expected # of rounds to achieve another win?

$$r_W = \mathbb{E}[R \mid W]$$

$$r_L = \mathbb{E}[R \mid L]$$



$$r_W = (.8)1 + (.2)(1 + r_L)$$



$$r_L = .3(1) + .7(1 + r_L)$$

$$r_L = .3 + .7 + .7r_L$$

$$.3r_L = .10$$

$$r_L = \frac{10}{3}$$

$$r_w = (.8)1 + (.2)(1 + r_L)$$

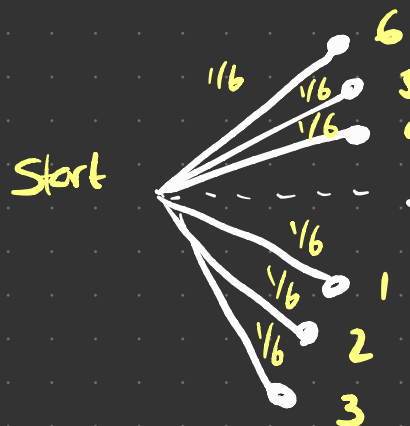
$$r_w = .8 + .2 + .2\left(\frac{10}{3}\right)$$

$$\approx 1.667 = \frac{5}{3}$$

$$\mathbb{E}[D] = \sum_{a \in A} a\,p(a)$$

We can choose to reroll!

$$= \frac{1}{6}(1+2+3+4+5+6) = 3.5$$



$$\mathbb{E}[D \mid R_1 > 3.5] = (6+5+4)\frac{1}{6} = \frac{15}{6}$$

$$\mathbb{E}[D \mid R_1 < 3.5] = \frac{1}{6}(3.5)(3) = \frac{1}{2}(3.5) = 1.75$$

$$\mathbb{E}[D] = \frac{15}{6} + 1.75$$

$= 2.5 + 1.75 = 4.25$

Consider Arbitary Dependency

Suppose the probability of winning the next round of the game now depends on the last 3 states ...

$$\underline{2} \cdot \underline{2} \cdot \underline{2} = 8 = 2^3$$

$$2^{10} = 1024$$

Defn: State

A state is a description of the system containing information needed to model its future evolution.

Markov Property

A stochastic process satisfies the Markov property when, given the present state, its next state is conditionally independent of path history

$$P(X_{n+1}=j \mid X_n=i_1, \dots, X_0=i_0)$$

$$= P(X_{n+1}=j \mid X_n=i)$$

Defn: Markov Chain

A <u>discrete-time</u> Markov Chain is a stochastic process

$$\{X_n : n=0,1,2,\dots\}$$

takes on values in a finite or countably infinite state space and satisfies

The Markov Property.

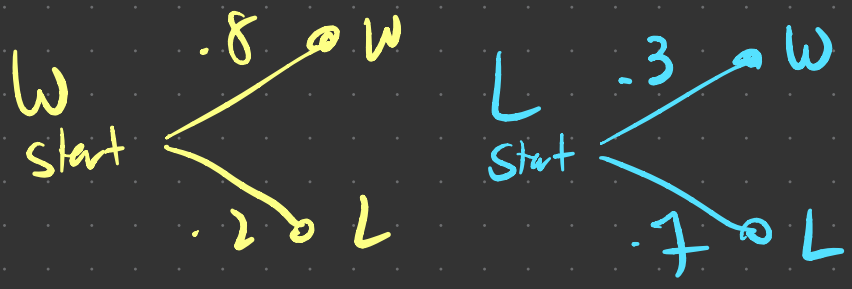
For a time homogeneous Markov Chain

$$P_{ij} = P(X_{n+1} = j \mid X_n = i)$$

$$P_{WW} = P(X_{n+1} = W \mid X_n = W)$$

$$P_{WW} = .8 \qquad P_{LL} = .7$$

$$P_{WL} = .2 \qquad P_{LW} = .3$$



Transition Matrix

$$P = \begin{array}{c} \\ W \\ L \end{array} \begin{array}{c} \begin{array}{cc} W & L \end{array} \\ \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix} \end{array}$$

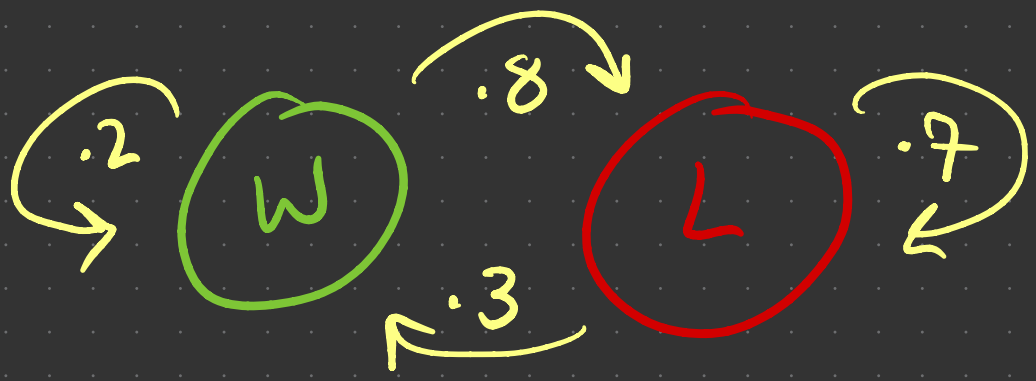
↳ stochastic matrix

$$\sum_j P_{ij} = 1$$

for game above of state order (W, L)

	WW	LW	WL	LL
WW →	·			0
WL →			·	
LW →		·		
LL →				·

$$P = \begin{array}{c} \\ W \\ L \end{array} \begin{array}{c} \begin{array}{cc} W & L \end{array} \\ \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix} \end{array}$$



ex) Suppose the current state is W

$$P(W_2 \mid W) = ?$$

$W \xrightarrow{.2} L \xrightarrow{.3} W \quad .06$

$W \xrightarrow{.8} W \xrightarrow{.8} W \quad \rightarrow .64$ +

$= .7$

$$P^2 = \begin{matrix} \\ W \\ L \end{matrix}\begin{matrix} \begin{matrix} W & L \end{matrix} \\ \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix} \end{matrix}^2$$

$$= \begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix}\begin{pmatrix} .8 & .2 \\ .3 & .7 \end{pmatrix}$$

$$= \begin{matrix} \\ W \\ L \end{matrix}\begin{matrix} \begin{matrix} W & L \end{matrix} \\ \begin{pmatrix} .7 & .3 \\ .45 & .55 \end{pmatrix} \end{matrix}$$

$$= P_{ij}^{(2)} = P_{WW}^{(2)} = .7$$

$$P_{WL}^{(2)} = .3$$

$$P_{LW}^{(2)} = .45$$

$$P_{LL}^{(2)} = .55$$

Defn: Chapman-Kolmogorov Equations

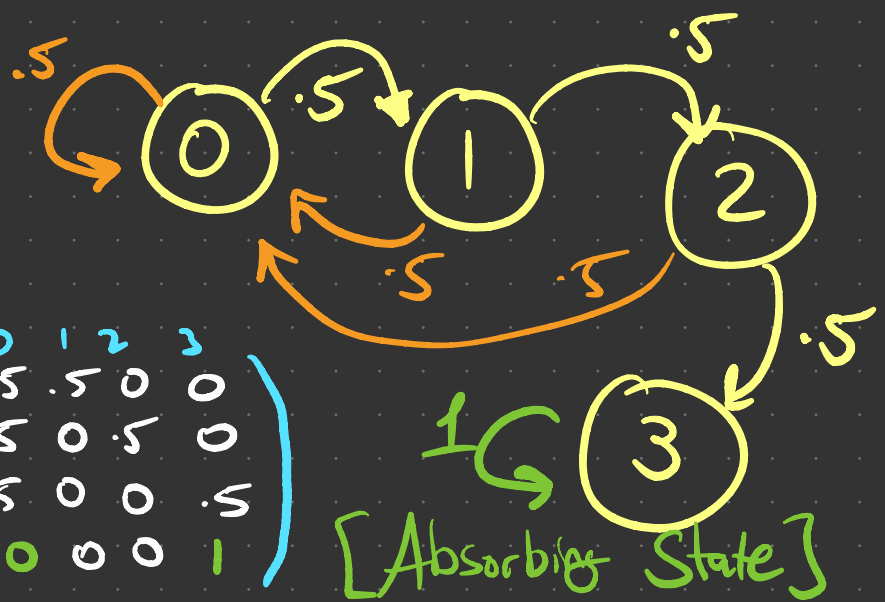
$$P_{ij}^{(m+n)} = \sum_K P_{i,K}^{(m)} P_{K,j}^{(n)}$$

$$\Rightarrow P^{(n)} = P^n$$

Markov Chains + Interview Question

Q: What is the probability of observing a chain of 3H in 10 fair coin flips?

$$S = \{0H, 1H, 2H, 3H\}$$



$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} .5 & .5 & 0 & 0 \\ .5 & 0 & .5 & 0 \\ .5 & 0 & 0 & .5 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

$P^{10} \rightarrow P_{0,3}^{(10)}$